

ROCKY MOUNT WILSON, NC, USA

WMO: 723068

Lat: 35.855N

Lon: 77.893W

Elev: 49

StdP: 100.74

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 93759

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.1	-4.8	-16.2	0.9	-2.6	-13.7	1.1	-1.5	9.5	8.0	8.7	7.0	2.2	310	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	34.8	24.7	33.6	24.4	32.4	24.0	26.3	32.0	25.6	31.1	25.1	30.2	3.5	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	19.9	28.5	24.1	19.1	27.7	23.7	18.6	27.2	81.5	31.8	78.7	31.3	76.4	30.3	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.4	6.5	DB	-10.7	37.3	2.6	1.7	-12.6	38.5	-14.1	39.4	-15.6	40.4	-17.5	41.6	(4)
(5)			WB	-11.7	27.3	2.4	0.7	-13.4	27.8	-14.7	28.3	-16.1	28.7	-17.8	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	5.1	6.8	10.5	16.1	20.5	24.8	26.4	25.7	22.6	16.6	10.6	7.0	(6)	
	DBStd	8.72	5.92	5.24	5.29	4.69	3.90	2.72	2.21	2.28	3.00	4.48	4.72	5.27	(7)	
	HDD10.0	534	176	117	60	6	0	0	0	0	0	4	48	124	(8)	
	HDD18.3	1784	412	324	247	98	25	1	0	0	4	87	234	352	(9)	
	CDD10.0	2770	22	28	77	189	327	445	509	487	379	208	67	32	(10)	
	CDD18.3	977	1	2	5	30	93	196	251	229	133	33	3	2	(11)	
	CDH23.3	8907	1	8	48	293	848	1886	2467	2083	1004	247	18	3	(12)	
	CDH26.7	3624	0	0	5	69	287	821	1135	909	343	55	1	0	(13)	
(14) Wind	WSAvg	2.6	2.8	3.0	3.2	3.2	2.6	2.4	2.2	1.9	2.3	2.4	2.5	2.5	(14)	
(15) Precipitation	PrecAvg	1157	92	81	104	74	92	102	122	119	124	81	75	83	(15)	
	PrecMax	1597	182	193	182	142	172	250	195	260	692	173	171	154	(16)	
	PrecMin	798	22	16	33	25	34	48	63	26	7	0	12	26	(17)	
	PrecStd	173	41	38	41	30	36	58	41	60	123	42	45	33	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.3	24.6	27.4	30.9	33.3	36.4	36.8	36.3	33.5	30.6	26.0	23.4	(19)	
		MCWB	17.6	17.8	17.4	20.6	22.5	24.5	25.5	24.7	23.3	23.3	19.7	19.1	(20)	
	2%	DB	19.7	21.4	24.5	28.6	31.3	34.1	34.9	34.1	31.9	28.0	23.1	21.1	(21)	
		MCWB	16.3	15.6	16.6	19.2	21.8	23.9	25.3	24.8	23.3	21.1	17.2	17.7	(22)	
	5%	DB	17.4	18.8	22.3	27.0	29.9	32.7	33.6	32.7	30.3	26.2	21.3	18.7	(23)	
		MCWB	14.1	14.3	15.4	18.6	21.3	23.6	24.8	24.5	22.5	20.2	17.2	16.2	(24)	
	10%	DB	14.7	16.6	19.8	24.7	28.0	31.2	32.3	31.4	28.7	24.1	19.0	16.3	(25)	
		MCWB	11.5	12.4	14.2	17.6	20.7	23.0	24.5	23.9	22.1	19.5	15.4	13.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.8	19.0	19.5	22.7	24.2	25.9	27.4	27.0	25.7	24.7	21.3	20.6	(27)	
		MCDB	20.8	22.8	24.0	27.2	29.6	31.9	33.7	32.3	30.3	27.3	22.7	22.0	(28)	
	2%	WB	16.8	17.5	18.3	21.0	23.2	25.2	26.4	26.2	24.9	23.3	19.5	18.4	(29)	
		MCDB	19.1	20.0	22.2	26.1	28.6	31.3	32.5	31.6	29.1	26.1	21.9	20.0	(30)	
	5%	WB	14.7	15.4	17.2	19.9	22.5	24.6	25.7	25.6	24.1	22.0	17.9	16.6	(31)	
		MCDB	16.8	18.5	20.3	24.7	27.6	30.5	31.5	30.6	27.7	24.5	20.1	18.4	(32)	
	10%	WB	12.1	12.6	15.7	18.7	21.6	24.0	25.2	24.9	23.4	20.5	16.3	14.0	(33)	
		MCDB	13.9	15.7	19.2	23.0	26.3	29.5	30.6	29.5	26.5	23.2	18.9	15.9	(34)	
(35) Mean Daily Temperature Range	MDBR		10.8	11.7	12.5	13.0	12.0	11.4	11.0	10.7	10.7	11.7	12.4	11.0	(35)	
	5% DB	MCDBR	13.7	14.2	14.8	14.7	13.6	12.9	12.5	12.4	12.7	13.2	14.0	12.6	(36)	
		MCWBR	10.1	9.9	8.1	6.6	5.5	4.5	4.3	4.4	5.1	6.7	8.7	9.5	(37)	
	5% WB	MCDBR	12.4	13.4	12.5	12.1	11.5	11.2	11.2	10.6	9.6	10.0	10.8	11.3	(38)	
		MCWBR	10.4	10.7	8.4	6.2	5.1	4.4	4.4	4.4	4.3	5.6	7.9	9.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.312	0.329	0.357	0.406	0.455	0.504	0.535	0.529	0.417	0.367	0.328	0.312	(40)	
	taud		2.500	2.449	2.404	2.269	2.166	2.075	2.004	2.010	2.324	2.443	2.506	2.535	(41)	
	Ebn at Noon		890	912	912	879	836	793	764	760	838	857	862	867	(42)	
	Edh at Noon		82	97	111	133	150	164	175	171	119	97	81	75	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.55	3.25	4.27	5.41	5.98	6.39	6.07	5.48	4.53	3.67	2.81	2.24	(44)	
	RadStd		0.18	0.36	0.30	0.38	0.45	0.41	0.33	0.30	0.36	0.47	0.22	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+190	+147
Regional (65 neighbors)		+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+106

Nomenclature: See separate page